less heat and fewer glares and lasts longer even though it is very expensive.

b) **Incandescent:** It is most commonly used, less expensive lighting produced by filament bulbs. It consumes more electricity and produces more glare and shadows. Incandescent lighting may also be used to provide heat for the office building.

c) **LED:** The light-emitting diode (LED) is one of today's most energy-efficient and rapidly developing lighting technologies. Quality LED light bulbs last longer, and are more durable. They also offer comparable or better light quality than other types of lighting.

d) **Solar Lamp:** A solar lamp also known as solar light or solar lantern, is a lighting system composed of an LED lamp, solar panels, battery, charge controller and there may also be an inverter. The lamp operates on electricity from batteries, charged through the use of solar photovoltaic panel.

Do You Know?

- Lighting a typical office overnight wastes enough energy to heat 1,000 cups of water for tea.
- LED light significantly reduces bacteria when compared to halogen or fluorescent bulbs.
- 95% of LED are completely recyclable

Types of Office Lighting System:

The lighting system consists of the following five types

1. Direct Lighting:

Approximately 90% to 100% of the light is directed downward to the working surface in this type of lighting. This system uses some shade or reflector for the source of light. It provides a good intensity of light. At the same time, it casts dark shadows around and raises direct and reflected glare that could cause eyestrain to the office employees.

2. Indirect Lighting:

In this system, 90% to 100% of the light is directed towards the ceiling or walls and then the directed light is diffused and reflected on the work surface. This type of lighting system is soft and avoids shadows and glares and does not cause eyestrain. This type of lighting system is highly useful for general illumination but not for office work.

3. Semi-Direct Lighting:

In this type of lighting, a transparent or translucent shade is used which directs 60% to 80% of the light downward. The remaining light is directed upward and the directed light is reflected from the ceiling. This lighting system is an improvement of direct lighting. So, dark shadows on the ceiling and glare are eliminated in this system.

4. Semi-Indirect Lighting:

A transparent or translucent shade, bowl, cover or reflector is placed under the

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